Pranav Viswanathan

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Education

Northeastern University Boston, MA
Master of Science, Computer Science *Sept 2024 – May 2026*

Vellore Institute of Technology Chennai, India
Bachelor's in Technology, Computer Science and Engineering *2024*

Work Experience

NergyLive May 2023 – Jul 2023
SDE Intern

- Built real-time HVAC anomaly detection using isolation forests and statistical process control, processing 10K+ sensor readings/minute with sub-100ms latency in Python
- Deployed anomaly detection system into production with cross-functional teams, reducing downtime incidents and cutting maintenance costs by 15%; built Plotly dashboards enabling engineers to identify root causes 3x faster

Projects

CaduceusAI: Three-Tier Medical AI Platform | *Python, FastAPI, Next.js 14, PostgreSQL, Redis, Docker, Ollama* Mar 2026 – April 2026

- Architected local-first three-tier medical AI platform (patient intake, clinical decision support, post-care) with on-device Ollama LLM inference, async httpx with 120s timeout, and automatic rule-based fallback for drug interaction detection and symptom triage
- Engineered clinician feedback loop draining Redis override/flag events into a JSONL retraining buffer designed to feed a LoRA fine-tuning pipeline via Hugging Face PEFT, with versioned assessments retaining full prompt-response history for training data reconstruction
- Secured platform with AES-256 Fernet PHI encryption, cookie-first HS256 JWT auth, bcrypt hashing, append-only audit logging, and SlowAPI rate limiting; orchestrated via Docker Compose with Alembic migrations

OmniRAG | *Python, FastAPI, LlamaIndex, LanceDB, Next.js, React, Docker, Ollama* Feb 2026 – Mar 2026

- Built a local-first Retrieval-Augmented Generation system with a FastAPI backend and Next.js frontend, enabling fully private on-device LLM inference via Ollama with session-isolated vector persistence in LanceDB
- Implemented BGE cross-encoder reranking with all-MiniLM-L6-v2 SentenceTransformer embeddings, improving retrieval precision through cosine similarity scoring across semantically chunked document collections
- Designed smart fallback pipeline routing low-confidence queries to an async web scraper (BeautifulSoup4, HTTPX, Wikipedia API) and built an interactive Semantic Knowledge Graph using react-force-graph-2d for cross-document relationship visualization with live citation tracking

Rebalance-AI: BlueBikes Demand Prediction | *Python, Apache Airflow, MLflow, Docker, GitHub Actions* Sept 2025 – Dec 2025

- Engineered end-to-end ML pipeline processing 3.16M ride records with Apache Airflow orchestration, implementing automated feature engineering for weather, temporal, and geospatial data
- Developed production-grade ensemble models (XGBoost, LightGBM, Random Forest) achieving $R^2 > 0.96$, with MLflow integration for experiment tracking, model versioning, and deployment automation
- Built CI/CD pipeline using GitHub Actions for automated testing and Docker containerization, with comprehensive monitoring tracking model performance metrics across 400+ bike stations

HawkEye | *Python, OpenCV, PyTorch, Docker, Shell Scripting* Sept 2025 – Nov 2025

- Built real-time multi-sensor fusion pipeline processing synchronized RGB, thermal, and LiDAR streams at 14-16 FPS with sub-100ms end-to-end latency, implementing modular sensor adapters for independent stream ingestion and frame alignment
- Implemented Kalman Filter-based multi-object tracking with IoU-based Hungarian assignment, maintaining consistent track IDs across 400+ frames with automated state prediction, gating thresholds, and track lifecycle management
- Developed configurable sensor degradation simulation framework modeling fog attenuation, Gaussian noise injection, and signal jamming across all three modalities, enabling systematic robustness evaluation under 10+ edge-case scenarios

Technical Skills

Languages: Python, Java, Go, JavaScript, TypeScript, SQL, Shell Scripting
AI & Machine Learning: RAG, LLM Fine-Tuning (LoRA/PEFT), Vector Embeddings, Cross-Encoder Reranking, Supervised/Unsupervised Learning, Ensemble Methods, Feature Engineering, Anomaly Detection
Frameworks & Libraries: FastAPI, Next.js, React, PyTorch, Scikit-learn, XGBoost, LightGBM, LlamaIndex, LanceDB, OpenCV, Hugging Face Transformers, SentenceTransformers, Pandas, NumPy
MLOps & Infrastructure: Docker, Apache Airflow, MLflow, GitHub Actions, Redis, Terraform, Prometheus, Git, Linux, pytest, Locust
Databases & Cloud: PostgreSQL, LanceDB, AWS (S3, EC2, RDS, Lambda, Fargate, SNS/SQS), Google Cloud Platform